Calculus I

Tutorial #4 (21–25 Oct 2024)

Topics:

- Rate of change
- Derivatives of trigonometrical functions
- The Chain Rule
- Implicit differentiation
- Related rates
- Linearization

Problem 1. An airplane takes off from an airport and lands an hour later at another airport, 400 miles away. If t represents the time in minutes since the plane has left the terminal building. Let $x(t)$ be the horizontal distance traveled and $y(t)$ be the altitude of the plane.

a) Sketch a possible graph of $x(t)$.

b) Sketch a possible graph of $y(t)$.

c) Sketch a possible graph of the ground speed.

d) Sketch a possible graph of the vertical velocity.

Problem 2. *(Particle motion)* At time t, the position of a body moving along the s-axis is $s = t^3 - 6t^2 + 9t$.

a) Find the body's acceleration each time the velocity is zero.

b) Find the body's speed each time the acceleration is zero.

c) Find the total distance traveled by the body from $t = 0$ to $t = 2$.

Problem 3. *(Bacterium population)* When a bactericide was added to a nutrient broth in which bacteria were growing, the bacterium population continued to grow for a while, but then stopped growing and began to decline. The size of the population at time t (hours) was $b = 10^6 + 10^4 t - 10^3 t^2$. Find the growth rates at

a) $t = 0$ hours.

b) $t = 5$ hours.

c) $t = 10$ hours.

Problem 4. *(Airplane takeoff)* Suppose that the distance an aircraft travels along a runway before takeoff is given by $D = (10/9)t^2$, where D is measured in meters from the starting point and t is measured in seconds from the time the brakes are released. The aircraft will become airborne when its speed reaches 200 km/h. How long will it take to become airborne, and what distance will it travel in that time?

Problem 5. Is there a value of c that makes

$$f(x) = \begin{cases} \dfrac{\sin^2 3x}{x^2}, & x \neq 0, \\ c, & x = 0, \end{cases}$$

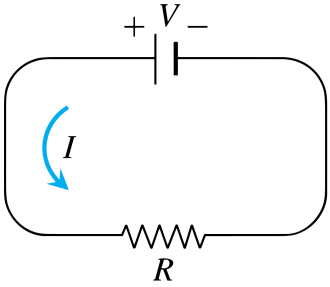
continuous at $x = 0$? Justify your answer.

Problem 6.

a) If $x^3 + y^3 = 16$, find the value of d^2y/dx^2 at the point $(2, 2)$.

b) If $xy + y^2 = 1$, find the value of d^2y/dx^2 at the point $(0, -1)$.

Problem 7. *(Changing voltage)* The voltage V (volts), current I (amperes), and resistance R (ohms) of an electric circuit like the one shown here are related by the equation $V = IR$. Suppose that V is increasing at the rate of 1 volt/sec while I is decreasing at the rate of 1/3 amp/sec. Let t denote time in seconds.



a) What is the value of dV/dt?

b) What is the value of dI/dt?

c) What equation relates dR/dt to dV/dt and dI/dt?

d) Find the rate at which R is changing when $V = 12$ volts and $I = 2$ amp. Is R increasing, or decreasing?

Problem 8. *(Distance)* Let x and y be differentiable functions of t and let $s = \sqrt{x^2 + y^2}$ be the distance between the points $(x, 0)$ and $(0, y)$ in the xy-plane.

a) How is ds/dt related to dx/dt if y is constant?

b) How is ds/dt related to dx/dt and dy/dt if neither x nor y is constant?

c) How is dx/dt related to dy/dt if s is constant?

Problem 9. *(Changing dimensions in a rectangle)* The length l of a rectangle is decreasing at the rate of 2 cm/sec while the width w is increasing at the rate of 2 cm/sec. When $l = 12$ cm and $w = 5$ cm, find the rates of change of

a) the area,

b) the perimeter,

c) the lengths of the diagonals of the rectangle.

Which of the above quantities are decreasing, and which are increasing?

Problem 10. Find the linearization of f at $x = a$. Then graph the linearization of f together.

$$f(x) = \sin x \text{ at } \quad \text{(a) } x = 0, \qquad \text{(b) } x = \pi.$$

Problem 11. The radius of a circle is increased from 2.00 to 2.02 m.

a) Estimate the resulting change in area.

b) Express the estimate as a percentage of the circle's original area.

Problem 12. *(An osculating circle)* Find the values of h, k, and a that make the circle $(x - h)^2 + (y - k)^2 = a^2$ tangent to the parabola $y = x^2 + 1$ at the point $(1, 2)$ and that also make the second derivatives d^2/dx^2 have the same value on both curves there. Circles like this one that are tangent to a curve and have the same second derivative as the curve at the point of tangency are called osculating circles (from the Latin *osculari*, meaning "to kiss").

THE END

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Answer to selected problems

4. 25 (sec) and ≈ 694.44 (m) 5. $c = 9$

10. $L(x) = x$ and $L(x) = -x + \pi$ 12. $\left(-4, \frac{9}{2}, \frac{5\sqrt{5}}{2}\right)$.